Camerala: A Privacy-Preserving Web Framework for Unobtrusive Longitudinal Context-Aware Monitoring

Kotaro Furukawa, *Student Member, IEEE*

***Abstract*—Ubiquitous affective computing faces critical bottlenecks: protecting user privacy in domestic spaces and correctly interpreting physiological signals across shifting contexts. We propose Camerala, a privacy-preserving web sensing framework for unobtrusive longitudinal monitoring. By executing remote photoplethysmography (rPPG) and facial landmark tracking entirely within the browser via WebAssembly, Camerala eliminates raw video transmission, solving the "privacy bottleneck." To evaluate its deployability in the wild, we conducted a 10-session autoethnographic proof-of-concept study comparing physiological reactivity between a University Laboratory and a private Home setting. The system successfully captured "Invariance Breaking": a cognitive stressor induced sympathetic mobilization (freezing, accelerated reaction times) in the Lab, but this autonomic response vanished in the Home. This demonstrates Camerala's robust edge-computing capabilities and highlights the critical necessity of context-aware, environmentally parameterized models for next-generation pervasive affective systems.**

***Index Terms*—Web-based Sensing, Single-Case Experimental Design, Invariance Breaking, Remote Photoplethysmography (rPPG), Affective Computing, Ecological Validity, Context Dependency, Polyvagal Theory, Challenge and Threat, Privacy-Preserving AI.**

I. Introduction

The vision of Affective Computing, as originally articulated by Picard within the MIT Media Lab in the late 1990s [?], was to endow machines with the capacity to recognize, interpret, simulate, and potentially influence human emotions. For over three decades, this vision has catalyzed extensive interdisciplinary research spanning computer science, psychology, neuroscience, and biomedical engineering. The ability to automatically detect user states—ranging from frustration and stress to engagement and flow—promises to revolutionize human-computer interaction (HCI), enabling systems that adapt dynamically to the user's cognitive and emotional needs.

Early work in this domain was predominantly conducted in highly controlled laboratory environments. Researchers utilized medical-grade sensors—such as electrocardiograms (ECG) to measure heart rate variability (HRV), electroencephalograms (EEG) to monitor cortical asymmetry, and galvanic skin response (GSR) sensors to track sympathetic arousal. These pioneering studies successfully established the somatic markers of basic emotions, enabling the development of classifiers that could distinguish high-arousal states (e.g., anger, fear) from low-arousal states (e.g., sadness, relaxation) with remarkable accuracy. Indeed, within the sanitized, noise-free confines of a sound-proofed laboratory, with a participant wired to a stationary rig, we can predict acute stress with near-perfect fidelity.

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K. Furukawa is with the College of Transdisciplinary Sciences for Innovation, Kanazawa University, Kanazawa, Ishikawa 920-1192, Japan (e-mail: kotaro0530@stu.kanazawa-u.ac.jp).

However, as the field matures and transitions from the laboratory to "in-the-wild" applications [?], a fundamental disconnect has emerged between the capabilities of these models and the complexity of real-world human experience. This disconnect, which we term the "Contextual Gap," places the field at a critical juncture: we have successfully democratized the *sensors* to measure physiology anywhere (via smartwatches, rings, and now standard webcams), but we critically lack the *contextual models* to interpret these signals correctly outside the lab.

A. The Crisis of Contextual Rigidity and "Invariance Breaking"

Current affective computing systems, particularly those deployed in consumer wearables or smart environments, often operate on the implicit assumption of *Physiological Invariance*. In machine learning terms, this assumes that the conditional probability of an affective state Y given a physiological signal X, denoted as $P(Y|X)$, is stationary across all domains D. However, we argue that this mapping is fundamentally non-stationary and dependent on the environmental context E:

$$P(Y|X;E_{lab}) \neq P(Y|X;E_{home}) \quad (1)$$

This discrepancy leads to what we define as **"Invariance Breaking"**—the systematic failure of affective models when transferred between environments. A model trained to detect stress in a driving simulator may completely fail when applied to a user sitting in their living room, not because the sensor is broken, but because the underlying causal structure of the physiological response has shifted. The "stress" response in a laboratory—characterized by vigilance and sympathetic mobilization—may be physiologically distinct from "stress" at home, where safety cues damp the autonomic response.

B. Theoretical Framework: Contextual Reactivity

To address Invariance Breaking, we ground our design in the Biopsychosocial Model [?] and Polyvagal Theory [?], [?].

These frameworks suggest that identical cognitive stressors can elicit fundamentally different physiological responses based on environmental appraisal. In unfamiliar, evaluative environments (like a University Laboratory), a stressor triggers "Threat" (sympathetic mobilization, vasoconstriction, freezing). Conversely, a private, familiar environment (like a Home) acts as a "Safe Haven," engaging the ventral vagal complex to dampen sympathetic reactivity. Therefore, robust ubiquitous systems must dynamically parameterize their physiological baseline against the user's context.

C. Research Questions and Contributions

This paper introduces Camerala and investigates its feasibility for context-aware monitoring through an autoethnographic case study, bridging the gap between advanced psychophysiological theory and modern web engineering. We pose two primary Research Questions (RQs):

- **RQ1 (Framework Feasibility)**: Can a purely client-side, privacy-preserving web framework (Camerala) be reliably deployed in contrasting real-world environments without requiring specialized hardware?
- **RQ2 (Invariance Breaking Demonstration)**: As a proof-of-concept, can this framework capture the hypothesized context-dependent shifts in physiological reactivity ("Invariance Breaking") between an evaluative Laboratory setting and a private Home setting?

To answer these questions, we developed **Camerala**, a purpose-built framework for longitudinal, context-aware affective experimentation. Our contributions are threefold:

1) **Camerala Framework**: We present a robust, open-source architecture for client-side physiological sensing. By processing all video data locally within the browser (Edge Computing), Camerala solves the "Privacy Bottleneck" that hinders longitudinal video collection in private homes. We detail the implementation of rPPG and facial landmark tracking using WebAssembly for real-time performance.
2) **Autoethnographic Deployment Strategy**: We demonstrate the utility of autoethnographic case studies (via Single-Case Experimental Design) for ubiquitous systems evaluation. While large-N group studies are necessary for broad psychological generalizations, dense longitudinal sampling ($N = 1; T = 600$) serves as a powerful feasibility stress-test for capturing delicate context-dependencies in real-world scenarios.
3) **Proof-of-Concept Case Study on Contextual Buffering**: Through a 10-session deployment, we provide a concrete demonstration of "Invariance Breaking." We show that the system successfully detects behavioral and physiological signatures of "Threat" in the Lab, while capturing their attenuation at Home. This highlights the practical necessity for context-aware affective models in future scalable deployments.

The remainder of this paper is organized as follows: Section II reviews the state-of-the-art in remote sensing and context-awareness. Section III details the technical implementation of Camerala. Section IV describes our SCED methodology. Section V presents detailed statistical results. Section VI discusses the theoretical implications for ubiquitous computing, and Section VII concludes.

II. Related Work

A. Remote Physiological Sensing (rPPG)

Non-contact physiological measurement has evolved from Blind Source Separation (e.g., ICA [?]) to Model-Based optics (e.g., POS [?]), and recently to Deep Learning (e.g., DeepPhys [?]). While state-of-the-art CNNs offer high accuracy in controlled settings, they frequently suffer from "Domain Shift" when deployed in-the-wild, severely degrading in unstructured lighting or motion [?]. To ensure cross-context robustness and real-time client-side performance, Camerala adopts a physics-based approach, isolating the pulse signal robustly against rigid head motion.

B. Context-Awareness in Affective Computing

The concept of "Context" in computing was formalized by Dey (2001) as "any information that can be used to characterize the situation of an entity." In Affective Computing, however, the integration of context has been slow and superficial. We categorize existing approaches into three levels of sophistication:

1) Level 0: Context-Agnostic: The majority of commercial "Stress Detection" algorithms fall here. They apply a static threshold: $HR > 90 \rightarrow Stress$. This fails in virtually all real-world scenarios due to the physiological confound of physical activity (metabolic demand vs. psychological stress).

2) Level 1: Context-as-Feature: Modern multimodal systems append context data to the feature vector. For example, a classifier might take inputs $[HR; GSR; Location_{GPS}; Noise_{dB}]$. The model learns to adjust its decision boundary based on location. While an improvement, this approach treats context as just another noisy sensor input. It assumes that the relationship between context and physiology is linear and additive.

3) Level 2: Context-as-Moderator (Interactionist): Our work advocates for this theoretical level. We argue that context is not a feature; it is a *lens* that fundamentally alters the semantic meaning of physiological arousal. Based on the Interactionist Theory of Emotion, emotion is constructed from the collision of Core Affect (Arousal/Valence) and Conceptual Knowledge (Context). The *same* core affect (high arousal) becomes "Fear" in a dark alley but "Excitement" on a roller coaster.

C. Methodological Shifts: Large-N vs. SCED

The "Gold Standard" in HCI research has long been the Between-Subjects Randomized Controlled Trial (RCT), typically with $N = 20 \sim 30$. This design assumes that by averaging across many individuals, individual differences will cancel out as "noise," revealing the "true" population effect. However, in psychophysiology, individual differences are not noise; they are the dominant source of variance.

1) **The Baseline Problem**: Person A's resting HR might be 60 bpm; Person B's might be 90 bpm. Averaging them is meaningless.
2) **The Reactivity Problem**: Person A might be a "Vascular Responder" (BP spikes), while Person B is a "Cardiac Responder" (HR spikes).

Pooling these heterogeneous phenotypes often washes out significant effects, leading to Type II errors. Single-Case Experimental Design (SCED) [**?**] avoids this aggregation fallacy. By treating the individual as their own control and sampling them repeatedly (in our case, 600 trials), we achieve high statistical power to detect *idiographic* causal relationships.

D. Privacy-Preserving Edge AI

A major barrier to longitudinal affective computing is the "Privacy Paradox": users desire personalized services but are increasingly wary of surveillance. Current commercial solutions (e.g., Affectiva, Amazon Halo) rely on cloud-based inference, streaming raw video or audio to centralized servers. This architecture poses unacceptable risks for monitoring in intimate spaces like the home (e.g., bedrooms).

- **Cloud Architecture**: High latency, high privacy risk. Raw data leaves the device.
- **Edge Architecture**: Zero latency, zero privacy risk. Raw data is processed locally; only anonymous features (HR, RT) are stored.

Recent advances in WebAssembly (Wasm) and WebGPU have made high-performance computer vision possible in the browser. By compiling C++ libraries (OpenCV, MediaPipe) to Wasm, we can run state-of-the-art inference at native speeds (60 FPS) without a single byte of video data leaving the user's machine. Camerala leverages this paradigm to resolve the Privacy Paradox, enabling ethical, long-term monitoring of intimate states.

III. System Architecture: Camerala

Camerala (derived from "Camera" + "Laboratory") is a comprehensive, open-source web framework designed for high-temporal-resolution physiological experimentation. Unlike traditional cloud-based platforms (e.g., Pavlovia, Gorilla) that operate on a client-server architecture, Camerala employs a "Thick Client" (Edge Computing) philosophy. This architectural choice is driven by two critical non-functional requirements: **Privacy Preservation** and **Real-Time Latency**.

A. High-Level Architecture

The system enables a complete experimental loop entirely within the browser sandbox (see Fig. **??**). The architecture consists of three coupled modules running on the main JavaScript thread and dedicated WebWorkers:

1) **The Task Engine**: Responsible for stimulus rendering, trial logic, and reaction time measurement.
2) **The Vision Pipeline**: Responsible for ingesting the webcam stream, performing facial landmark inference, and extracting raw physiological signals.
3) **The Data Manager**: Responsible for fusing the asynchronous streams (Vision Frame Time vs. Task Event Time) and securing data persistence.

Fig. 1. Overview of the Camerala System Architecture. The system runs entirely client-side, processing webcam frames via MediaPipe Face Mesh (Wasm) to extract rPPG and Head Motion signals in real-time. Data is synchronized with the Psychophysics Task Engine (Canvas API) and stored locally, ensuring no raw video ever leaves the user's device.

B. Technical Implementation: The Vision Pipeline

We utilize the MediaPipe Face Mesh [**?**] solution, which provides 468 3D facial landmarks in real-time. This model is executed via TensorFlow.js with the WebGL backend to leverage the client's GPU acceleration.

1) Feature Extraction Logic: To quantify psychophysiological states, we extract two primary signals: *Head Motion (Freezing)* and *Exposure Fluctuation*.

1. Head Motion Index (M_t) The physiological "Freeze Response" (a defensive reaction to threat) translates to a